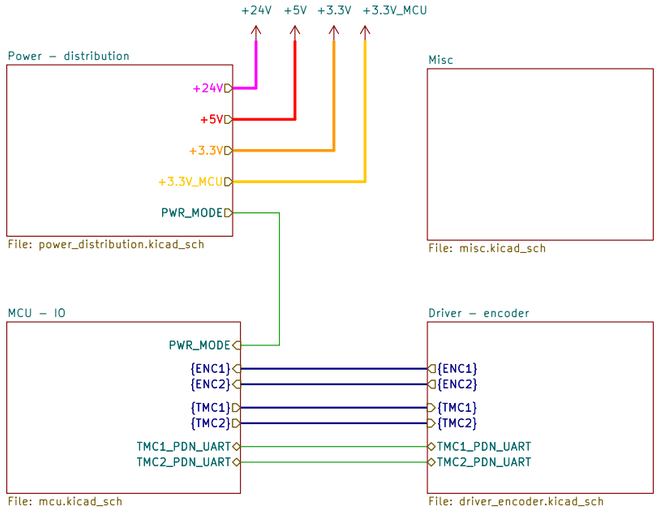
Dual Stepper Driver Evaluation Board

2026-10-22
Rev 1.0



To-Do:

During PCB design phase:
- HSE oscilator
- simulaties
- check USB speed
-> full speed no impedance control

Ordering:
- Check JLCPCB preview PCB

Final design:
- should split IO in seperate sheet
- IPC design rules
- block diagram
- firmware info? State diagram?

#	Test	Procedure	Pass criterion	Result
1	24V supply	24 V only, no motors	+5 V within 4.85–5.15 V, +3.3 V within 3.20–3.40 V, ripple below 50 mVpp	
2	USB supply	USB only, with a C–to–C cable and an A–to–C cable	Board is recognised over USB with both; drivers unpowered	
3	Supply switching	Unplug/replug 24 V while USB is connected, then unplug USB while 24 V is connected; scope +3.3V_MCU	No MCU reset, no USB disconnect, +3.3V_MCU stays above 3.0 V	
4	Reverse polarity	–24 V from a lab supply limited to 0.5 A, for 10 s	Draws under 1 mA; board works afterwards	
5	Hot–plug	Plug in the adapter with the 4 m cable 10 times; scope TP48	Peak below 29 V	
6	Brown–out	Ramp a lab supply 0 → 24 → 0 V slowly, motors connected	No motor movement; clean start and stop	
7	Motor enable	Power up, hold reset, flash firmware, all with 24 V present	No holding torque until the firmware enables the motors	
8	Driver UART	Write and read back GCONF and IHOLD_IRUN on both drivers	Both answer, values match, no error flags	
9	Motor current	Maximum set current; measure across the sense resistor (I = V / 0.22 Ω)	Within ±10 % of the setting and at most 1.0 A RMS	
10	Temperature	Both motors at maximum load for 30 min, board in its enclosure	No overtemperature warning, TMC2209 below 90 °C, other parts below 70 °C	
11	Braking energy	Hardest deceleration at maximum load, and cut 24 V while running	+24 V stays below 29 V	
12	Encoder count	100 revolutions forward and back at maximum speed	Back to exactly 0 counts; one index pulse per revolution; direction correct	
13	Lockstep	Both motors driving the ring gear	Difference between the two encoders below 20 counts	
14	Encoder fault	Unplug an encoder while running	No false counts; firmware flags the fault	
15	USB under load	Stream data for 30 min during maximum acceleration	0 errors, 0 disconnects	
16	Programming	Flash over SWD with the ST–LINK; enter DFU with BOOT0	Both work	
17	ESD	±4 kV contact discharge on the USB shell and the terminals	Keeps working or recovers by itself	

Standard passive component values

If not mentioned otherwise
R: tolerance=1%, package=0603
C: dielectric (not polarized)=X7R, package=0603

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for informational design notes.

DESIGN NOTE:
Example text for debug design notes.

ELECTRICAL CHARACTERISTICS:
Voltage supplied, Current drawn, ...

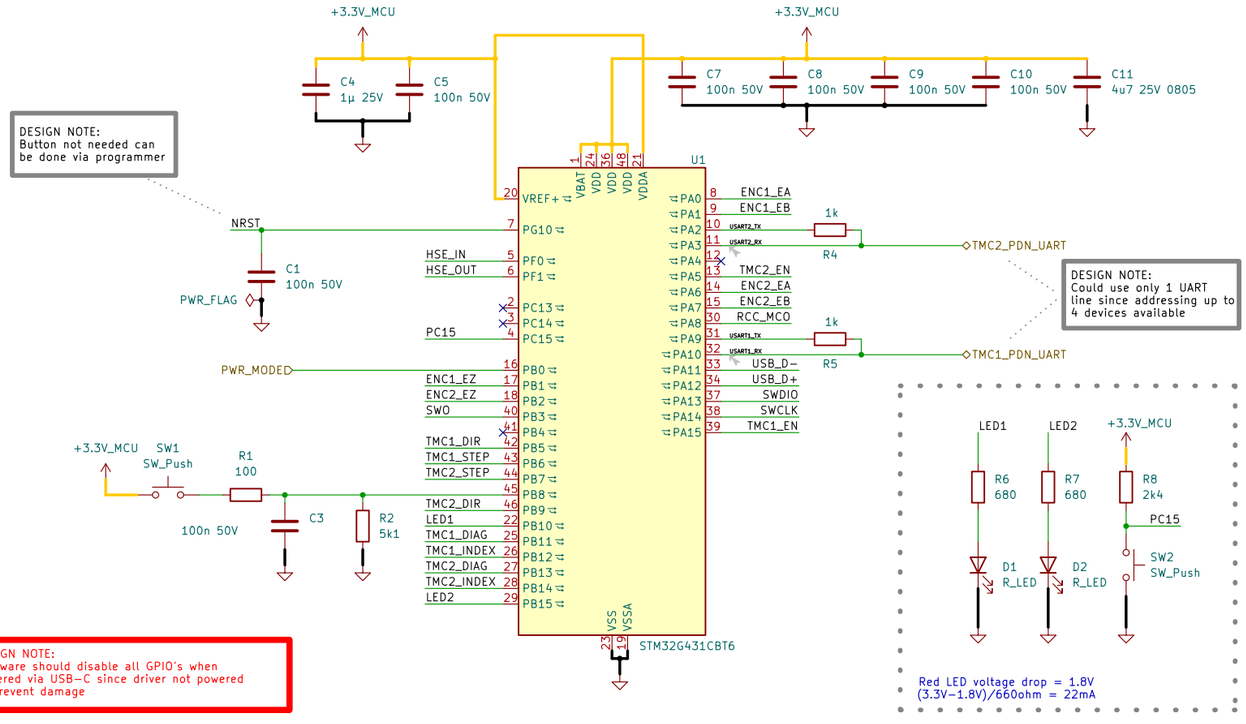
DESIGN NOTE:
Example text for critical design notes.

LAYOUT NOTE:
Example text for critical layout guidelines.

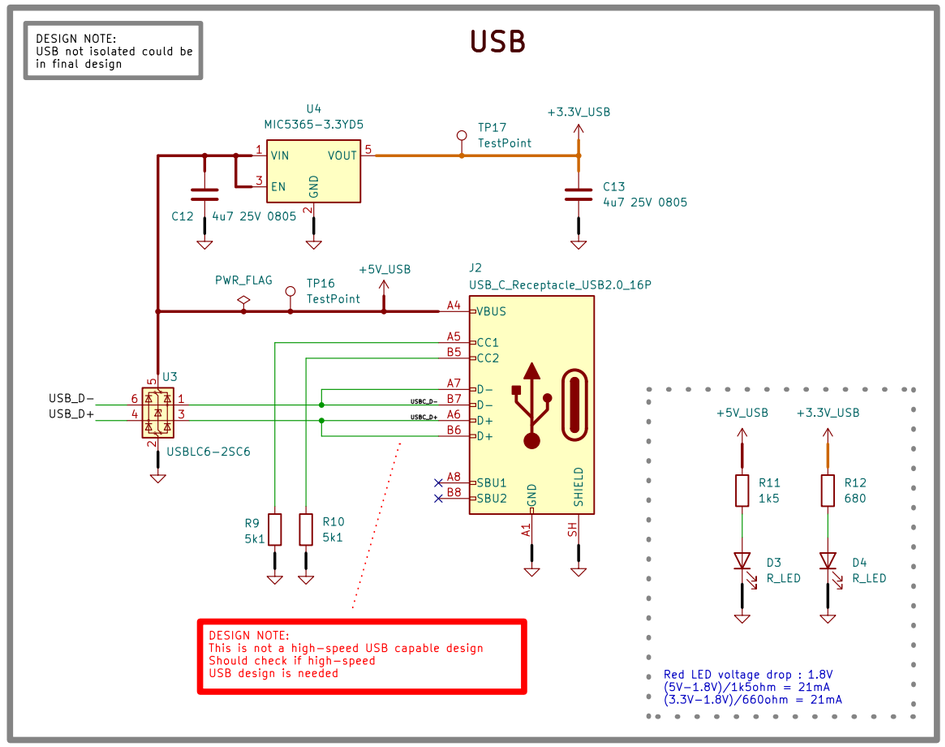
Sheet: /
File: stepper_motor_eval_board_v1.0.kicad_sch
Title: Stepper Motor Evaluation Board
Size: A3 | Date: 2026-10-23 | **Rev: 1.0**
KiCad E.D.A. 10.0.6 | Id: 1/5

ELECTRICAL CHARACTERISTICS:
3V3 30 mA
-> should be checked with stm tool

MCU



USB



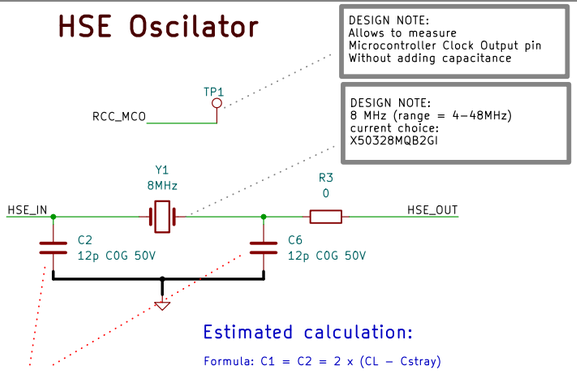
DESIGN NOTE:
No need of LSE oscilator since no sleep activity

DESIGN NOTE:
HSE oscilator since space and BOM allows it
Better accuracy timing for everything including UART and USB

DESIGN NOTE:
There are no constraints on crystal size, then it is recommended to choose a crystal with the smallest possible ESR.

DESIGN NOTE:
If there are no constraints on power consumption, it is recommended to choose crystals with high load capacitance.

HSE Oscilator



DESIGN NOTE:
Allows to measure Microcontroller Clock Output pin Without adding capacitance

DESIGN NOTE:
8 MHz (range = 4-48MHz) current choice: X50328MQB2GI

DESIGN NOTE:
Capacitors values should still be calculated depending on PCB traces etc.

Use 0402 footprint for easy replacement of caps when testing

Estimated calculation:

Formula: C1 = C2 = 2 x (CL - Cstray)

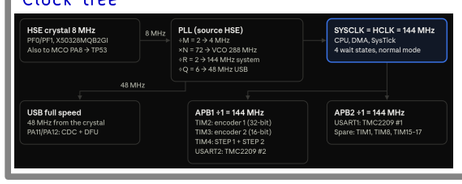
with Cstray estimated = 10pF p.100 datasheet MCU
C1 and C2 should be between 5pF-20pf

X50328MQB2GI:
CL = 16pF -> so that C1, C2 are in middle of allowed range
ESR = 80ohm
Tolerance = +-10ppm
Drive level = 100µW or higher

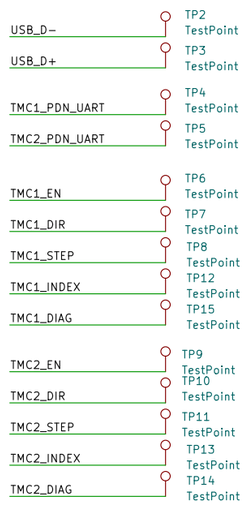
C1 = C2 = 12pF (estimate) -> good in middle of range

LAYOUT NOTE:
- place crystal and caps very close to OSC pins
- keep traces short and symmetrical
- avoid noisy digital routing near the oscillator loop
- avoid excessive vias in crystal traces
- respect the datasheet's ESR and drive recommendations

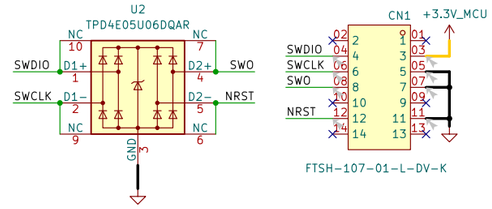
Clock tree



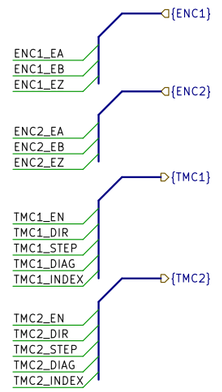
Testing points



SWD (Debug) Port



Interconnections



Sheet: /MCU - IO/
File: mcu.kicad_sch

Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 2/5

Stepper motor drivers

ELECTRICAL CHARACTERISTICS:
Driver current depending on motor: 1A/phase with 2ohm phase resistance
Pmotor_coil = (1A)² x (2ohm) = 2W
Motor has two coils so : 2W x 2 = 4W per motor
Idriver = P/U = 4W / 24V = 0.167A
Two drivers in total so : 0.167A x 2 = 0.33A

Two drivers = 24V 0.5A

One driver IO = 3V3 30µA
Two driver IO = 3V3 60µA

DESIGN NOTE:
Should still check layout example p82

DESIGN NOTE:
Do not power MCU output pin when IC not powered= when MCU only powered by USB

DESIGN NOTE:
Aluminium electrolytic capacitor

LAYOUT NOTE:
Place near IC with short path to die pad

DESIGN NOTE:
Chopper mode selection done via UART

DESIGN NOTE:
Diagnostic and StallGuard output.
Hi level upon stall detection or driver error.
Diagnostic done via UART

DESIGN NOTE:
In combination with a mechanical home switch, a more precise homing is enabled.
see p.8 and p66

DESIGN NOTE:
Removed osciloscope testpoints because can just measure at the connector

DESIGN NOTE:
Osciloscope testpoints for current sense through phases of stepper

LAYOUT NOTE:
Connect directly to GND plane and use low inductivity SMD type

DESIGN NOTE:
p.52 current sense resistor of 220mOhm for 1A motor

DESIGN NOTE:
Do not hot plug motor connector

{TMC1} TMC1_STEP TMC1_DIR TMC1_EN TMC1_DIAG TMC1_INDEX TMC1_PDN_UART

{TMC2} TMC2_STEP TMC2_DIR TMC2_EN TMC2_DIAG TMC2_INDEX TMC2_PDN_UART

+3.3V +24V

C15 100n 50V · C16 100n 50V · C17 100n 50V · C20 100n 50V · C21 100µ 50V · D6 SMAJ24A

C19 2.2µ 25V · C18 22n 50V

U7 TMC2209-LA

STEP DIR CLK PD/UART MS1/AD0 MS2/AD1 SPREAD EN STDBY DIAG INDEX VCC_IO VS VCP 5VOUT VREF CPO CPI OA1 OA2 OB1 OB2 BRA BRB GND NC

R13 10k · R14 680 · R15 220m 2010 · R16 220m 2010

J4 Motor1_Conn

TP31 TP32

Red LED voltage drop = 1.8V
(3.3V−1.8V)/660ohm = 22mA

D5 R_LED

C23 100n 50V · C24 100n 50V · C25 100n 50V · C28 100n 50V · C29 100µ 50V · D8 SMAJ24A

C27 2.2µ 25V · C26 22n 50V

U10 TMC2209-LA

R17 10k · R18 680 · R19 220m 2010 · R20 220m 2010

J6 Motor2_Conn

TP46 TP47

Red LED voltage drop = 1.8V
(3.3V−1.8V)/660ohm = 22mA

D7 R_LED

Encoder – RS422 differential signals to logic signals

ELECTRICAL CHARACTERISTICS
1 AM26LV32E :
3V3 17mA max supply current
2 AM26LV32E :
3V3 17mA x 2 = 34mA

ELECTRICAL CHARACTERISTICS:
1 Encoder :
5V 50mA estimated, to be measured
2 Encoders :
5V 50mA x 2 = 100mA

LAYOUT NOTE:
Should check to add termination resistance based on simulation

LAYOUT NOTE:
No need for testpoints on differential line since everything can be tested via connector

{ENC1} ENC1_EA ENC1_EB ENC1_EZ

{ENC2} ENC2_EA ENC2_EB ENC2_EZ

+3.3V +5V

C14 100n 50V · C22 100n 50V

TP18 TP19 TP20 TP33 TP34 TP35

U5 U8 AM26LV32EIDRG4

VDD GND 1A 1B 2A 2B 3A 3B 4A 4B 1Y 2Y 3Y 4Y G G

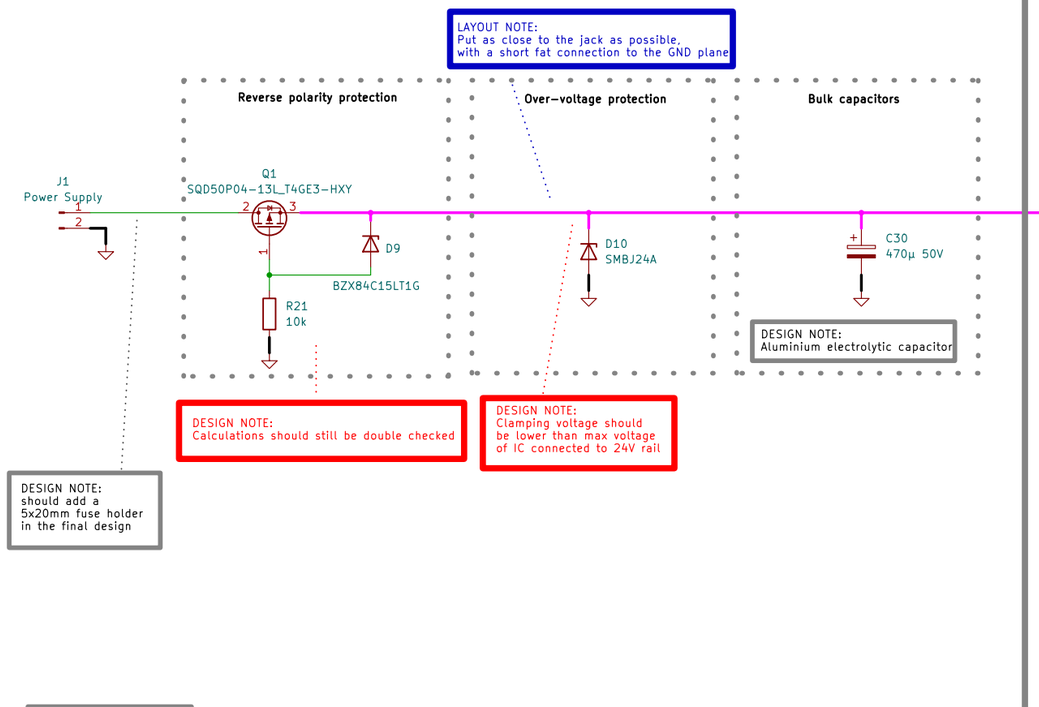
J3 Encoder1_Conn · J5 Encoder2_Conn

U6 U9 TPD6E05U06RVZR

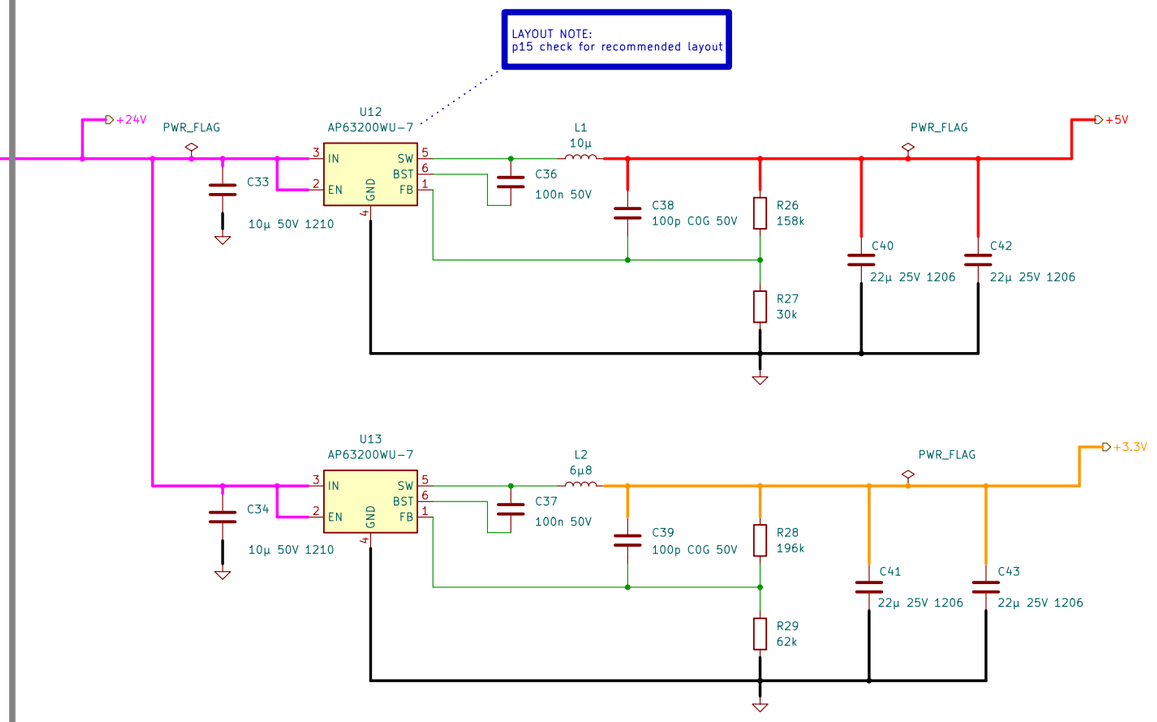
D1+ D1− D2+ D2− D3+ D3− GND NC

Sheet: /Driver – encoder/	
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Size: A3	Date:
KiCad E.D.A. 10.0.6	Id: 3/5

Power supply protection circuitry



DC-DC Buck converters



Current consumption

Voltage rail	Components	Estimated current draw	Dimensioned current draw
24V	2x motor (driver)	0.5A	2A
5V	2x encoders	0.1A	1A
3V3	2x driver IC, RS422 signal converter	60µA + 34mA = 0.03406A	0.5A
5V_USB	MCU	30mA	0.5A

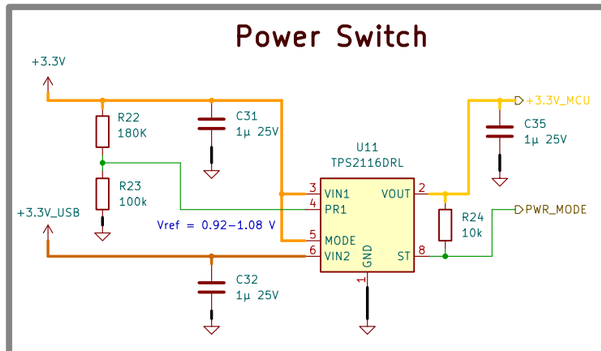
Component selector DC-DC buck converter

Table 1 shows a list of recommended component selections for common output voltages for the AP63200 and AP63201 referencing Figure 20.

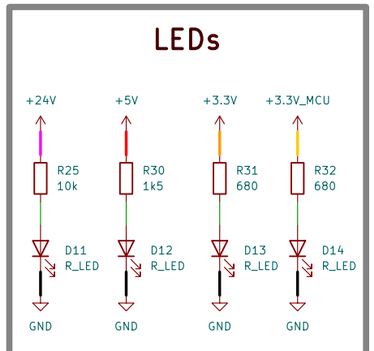
AP63200/AP63201							
Output Voltage (V)	R1 (kΩ)	R2 (kΩ)	L (µH)	C1 (µF)	C2 (µF)	C3 (nF)	C4 (pF)
1.2	30.9	62	2.2	10	2 x 22	100	100
1.5	54.2	62	2.2	10	2 x 22	100	100
1.8	77.5	62	3.3	10	2 x 22	100	100
2.5	131	62	3.3	10	2 x 22	100	100
3.3	196	62	6.8	10	2 x 22	100	100
5	157	30	10	10	2 x 22	100	100
12	249	18	10	10	2 x 22	100	56

Table 1. Recommended Component Selections for AP63200/AP63201

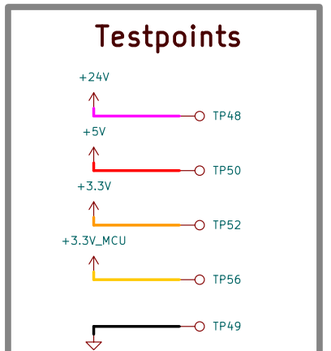
Power Switch



LEDs



Testpoints

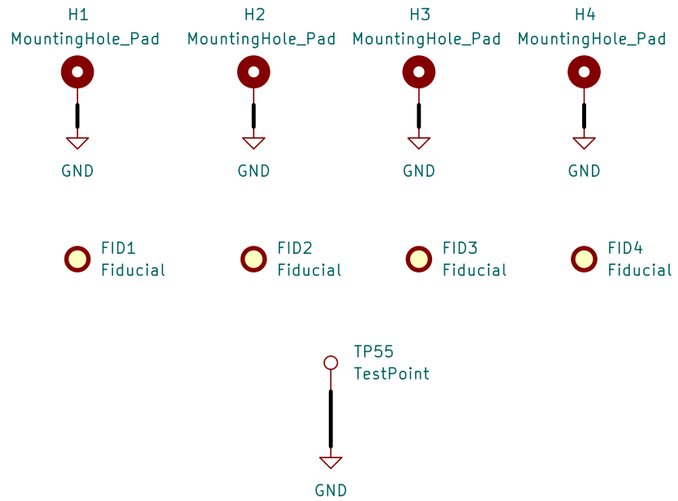


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Title:

Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 4/5

Misc



Sheet: /Misc/
File: misc.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 5/5